

Tunisia, audited — investor economics

The FX bridge by leg, per-pathway DSCR computed in code, the capital stack, and the exact signatures that move “commercially validated” from zero.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

1. The FX bridge, tiered by confidence

LEG	\$M/YR	GRADE
Strategic metals (U + heavy REE, acid stream)	\$75-86M	STRONG
Solar fuel substitution (500 MW)	\$95-116M	PLAUSIBLE
Hydrogen → ammonia (loop-internal offtake)	\$7-25M	SPECULATIVE
Terfas (desert truffles)	\$14-29M	PLAUSIBLE
Food import substitution	\$30-30M	SPECULATIVE
Compute colocation services	\$15-15M	SPECULATIVE
ESG financing delta	\$8-12M	PLAUSIBLE
Total (mid)	~\$279M/yr	tiered by grade below

\$80M carries a proven flowsheet or a cited anchor; \$137M is gated; \$61M rests on assumed volume or price. Zero dollars are commercially validated — the first buyer moves the ledger.

1.1 The Fed transmission — measured, not restated

World Bank API, measured 2026-09-09 (Pg1): external debt **\$40.46B**, **78.7% of GDP**; debt service **26.2% of export revenues**; reserves **\$9.34B** = 4.3 months of imports; food \$3.00B + fuel \$4.90B of imports; CA -1.5% of GDP; growth 2.5%, inflation 5.2% (2025).

Channel 1 — the debt trap: the “original sin” (Pg12) — no long-term borrowing in the dinar abroad — exposes the state to the Fed cycle (Pg13): Fed hikes lift global yields, refinancing turns expensive, and the 26.2% of export revenues that debt service absorbs compounds while reserves (4.3 months) drain.

Channel 2 — imported inflation: the rate differential pulls capital to US safe assets, the dinar depreciates, and the dollar invoice on food (\$3.00B) and fuel (\$4.90B) inflates. The BCT — autonomous, non-monetizing under Organic Law 2016-35 (Pg4/Pg5) — must hold rates to defend the dinar: correct orthodoxy, strangled growth.

The loop’s answer: owned export engines earn ~\$117M/yr of gross exports outside the Fed cycle; the substitution legs take \$152M/yr out of the dollar invoice directly; reserves rise, the rating runs B-/Caa1 → BB-, and the refinancing premium shrinks for the state with its own hard-currency engines. Not defiance of the Fed — independence from the channel.

The macro context is the thesis: the same Fed transmission that sets the emerging-market cycle prices this state’s debt, and the loop’s engines are the only line item in the country that earns hard currency *outside* that channel. That is why the FX bridge above is the investor’s first table, not an appendix.

2. Pathway economics — computed in code (nations_v5.py)

P2 · Strategic metals (acid-stream U + heavy REE)

LINE	VALUE
Revenue	\$75-86M/yr (716.5k lbs U&sb3;O&sb8; @ \$80-95/lb + ~585 t REO)
OPEX	\$54-74M/yr (SX \$59-83/lb + shared circuit)
Gross margin	0-37%
CAPEX / payback	~\$100M / 6.3 yr (mid)
DSCR (60% debt @7%/15y)	2.40 — passes at contract prices
Gate	pilot U recovery ≥85% · all-in ≤\$75/lb · offtake signed

P3 · Solar IPP (500 MW) — the honest finding

LINE	VALUE
Revenue / OPEX	\$27.2M / \$8M (500 MW × 20% CF × \$31/MWh)
CAPEX	~\$550M
DSCR @8%/20y	0.46 — fails

LINE	VALUE
DSCR @4.5% DFI/25y	0.69 — fails
Bankable region (DSCR ≥1.15)	$\$0.70/W + 21\% CF \cdot \$0.70/W + 22\% CF \cdot \$0.70/W + 23\% CF \cdot \$0.70/W + 24\% CF \cdot \$0.80/W + 24\% CF$
Clearing PPA	≥\$45652/MWh at mid-case capex/CF
Verdict	bankable ONLY lean, inside the region, or with grant/sovereign support — published, not hidden

P4 • Municipal desalination (150,000 m³/d)

LINE	VALUE
Revenue	\$30.1M @\$0.55 → \$38.3M @\$0.70 blended
OPEX / CAPEX	\$24.6M (@\$0.45/m³) / ~\$165M
DSCR @\$0.55	0.54 — fails
DSCR @\$0.70	1.36 — clears
Gate	≥60% contracted offtake · blended ≥\$0.68/m³ · brine permit

The water stack — line-by-line, with each credit's failure mode

The \$0.347/m³ audit: the base is measured anchors; each credit is a loop-internal yield with a named failure mode; the degraded cascade still clears the \$0.68 gate.

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The people's energy is electrons, not hydrogen — and the H&sb2; lives inside the loop

The loop's H&sb2; (2.63 kt/yr) is an industrial molecule: per kWh of heat, green H&sb2; at \$0.21 is 7× the loop's \$0.031/kWh electron; blending caps at 5-20% by volume (Pg20/Pg21/Pg22); the whole H&sb2; volume equals 0.0% of grid-gas demand. The molecule goes where only a molecule works: ammonia for the GCT chain, inside the loop. the electrolyzer sits on the desal line, and everything it touches stays in the loop: the water it splits is the SWRO permeate (0.04% of the 150,000 m³/d output); its un-split reject stream is clean water that returns to the product line; its oxygen by-product aerates the spirulina ponds and greenhouses; its stack heat (15–20% of input) joins the kiln and DC reject heat on the desal preheat circuit; and the minerals concentrated in its flush stream join the brine valorization line. The electrolyzer is not a consumer of the loop — it is another recovery point inside it.

3. The capital stack and the ask

Phase o: \$150-250k, 90 days, no construction — the price of replacing every projection with a measurement (itemized in the complete audit). **Then, module by module, on public gates:** \$58k health pilot · \$100M metals · \$550M solar · \$165M desal · \$5-30M truffles · \$2-5M food. Security per module: offtake assignment, STEG PPA, municipal contracts, MIGA/DFI channel — modeled, not committed.

What moves “commercially validated” from zero

GCT MOU + first assay batch · STEG PPA award · desal LOI ≥60% at ≥\$0.68/m³ · first truffle fruit + ≥€20/kg contract · 100-point water panel + 90-day health KPIs · 3,000-family cohort baseline. Each is a Phase-o/1 deliverable with a date.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

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